GCI World 2026 final assignment: Company A

Keeping customers past their first year

A data-driven retention engine for Company A: who is about to leave, why, and what keeping them is worth

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Executive summary

Company A can cut churn profitably by acting before the first anniversary

The first-year cliff

29% → 65%

churn rate from month 9 to month 11 of service

The early warning

64%

churn when monthly minutes fall by more than half, vs 44% if stable

Our model

0.706

ROC-AUC of LightGBM, the best of four models we tested

The value

+$28K

net per monthly campaign per 100,000 customers; a random list loses $27K

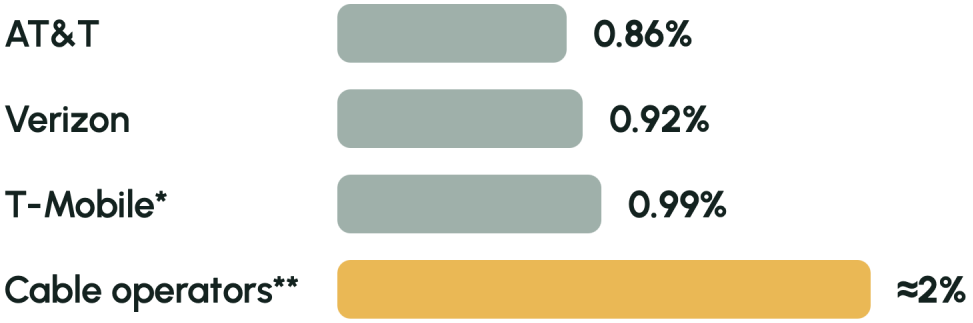
We recommend A 3-month pilot of a monthly retention engine, measured against a control group, then roll-out to all customers.

Churn rates: balanced study sample (49.6% churners). Money: re-scaled to a realistic 2% monthly churn.

US market context

US operators now grow by keeping customers

Monthly postpaid phone churn, Q2 2026



≈22% of customers lost per year at 2% a month, our base case for Company A

*Account churn, as T-Mobile reports it. **Comcast and Charter, analyst estimate. Sources: [1], [2].

Retention is the growth lever

Verizon's Q2 2026 gain came mostly from lower churn; promotions cost it about 1.9 points of margin [2].

The phone is the lock-in

About 95% of Verizon's postpaid contracts are month-to-month. Carriers hold customers with 36-month device payments and bill credits [1][3].

Analytics pays

Operators that manage their customer base with analytics have cut churn by up to 15% [4].

Sources: [1] TelecomLead 2026; [2] Light Reading 2026; [3] Fierce Network 2022; [4] McKinsey 2017

Company A data

We analysed 100,000 customers and 99 variables

100,000
customers: a profile file and a usage file, joined on customer ID

$58.72
average monthly bill per customer (median $48.20)

49.6%
left in this balanced study sample, so we compare groups, not levels

43 of 99
variables have gaps; household data is up to 49% empty

How we worked

1
Clean and merge
Join both files; flag 5 negative bills and 133 negative handset ages

2
Explore
Churn rate across 8 candidate drivers and 10 customer attributes

3
Model
Four models, 5-fold cross-validation, 18 new features

4
Explain
A main churn reason for every customer

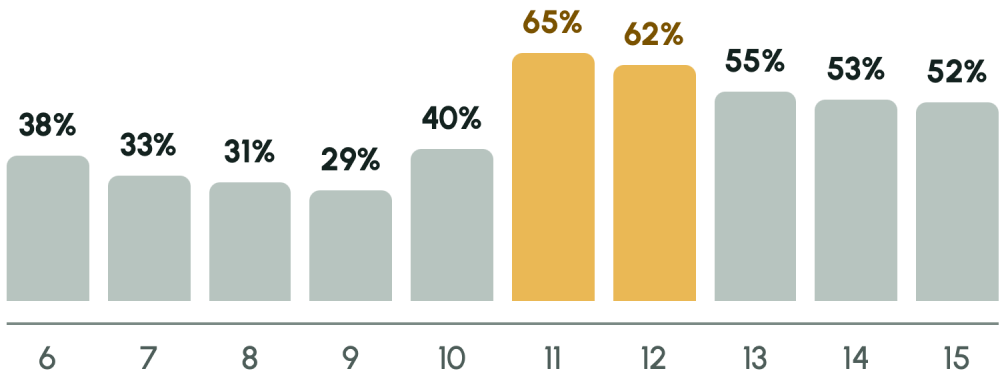
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Value
Campaign simulation at US churn levels

Data: Company A dataset (Client.csv, Record.csv). Code written with help from generative AI [9].

What the data shows: 1 of 3

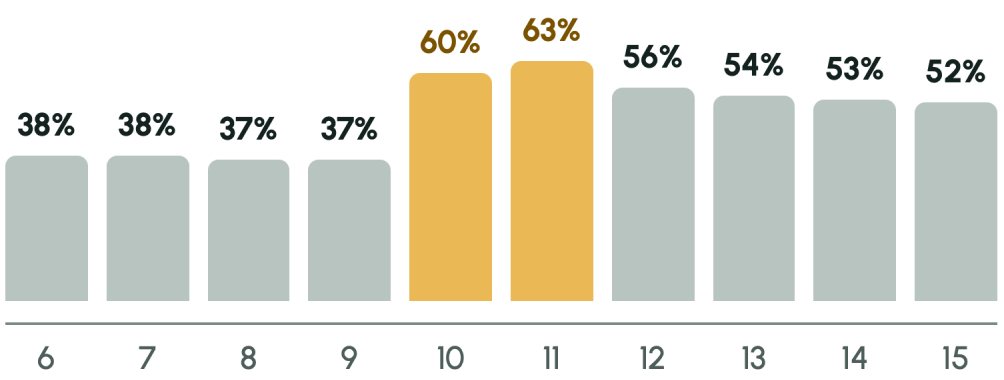
Churn spikes around the first anniversary

Churn rate by month in service



Month 9: 29%. Month 11: 65%. From month 13 on, churn settles near 50%.

Churn rate by age of handset, in months



9-month-old handset: 37%. At 10 months: 60%. Older handsets keep rising, to 58% past 2 years.

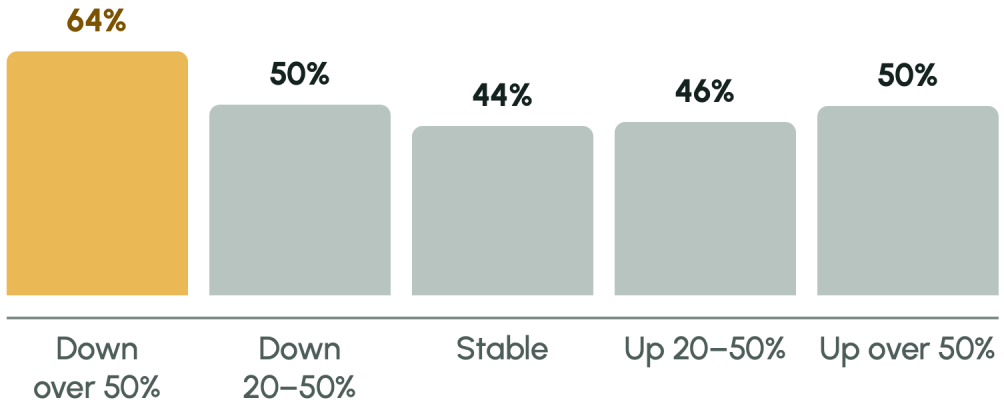
Likely cause: a one-year agreement or handset deal running out. The data has no contract-end field, so the pilot should confirm it. US carriers avoid this cliff with 36-month device plans [3].

Source: Company A data; groups of 300+ customers. Balanced sample: compare groups, not levels.

What the data shows: 2 of 3

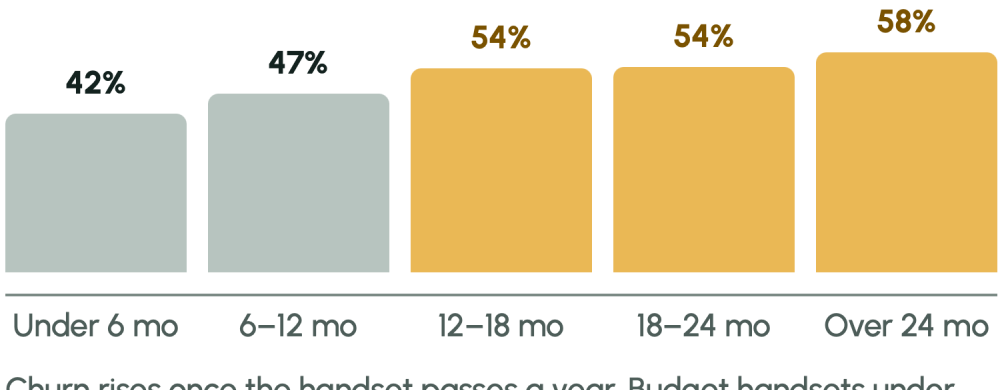
Falling usage and old handsets warn of churn

Churn rate by change in monthly minutes



Customers whose minutes halved churn at 64%, vs 44% for stable users: the most common warning sign.

Churn rate by age of current handset



Churn rises once the handset passes a year. Budget handsets under $40 churn at 57%, vs 45% at $130 and up.

So what: both signals sit in data Company A already collects every month, so they can trigger a check-in call or a handset offer before the customer leaves.

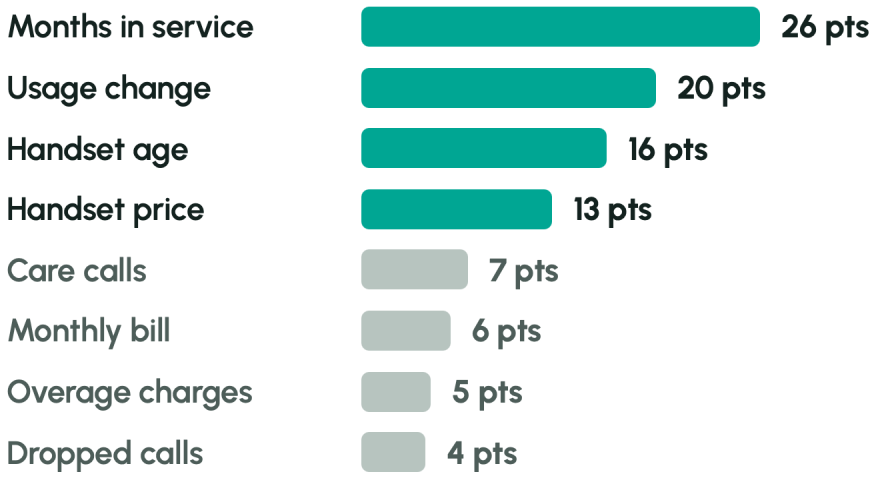
Source: Company A data. Usage change: latest month's minutes vs the previous 3-month average.

What the data shows: 3 of 3

What doesn't drive churn, and where the money is

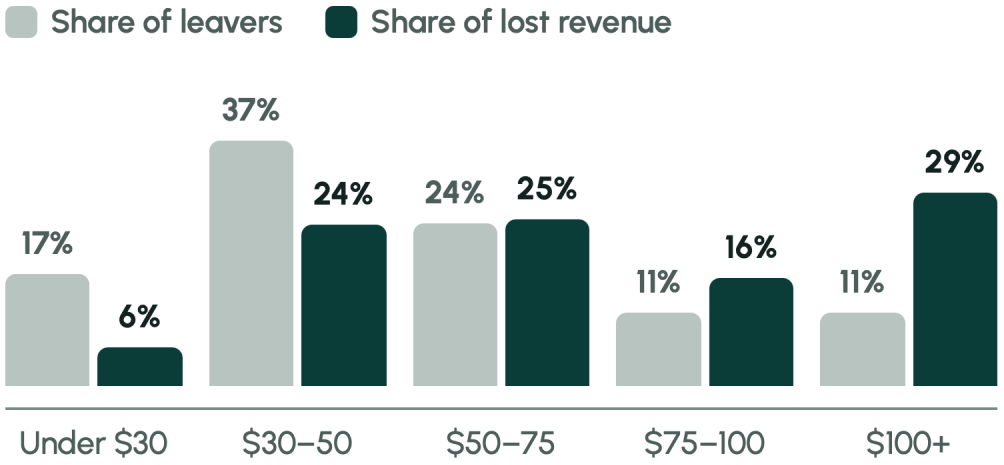
How much churn varies across groups

Gap between highest and lowest group, in percentage points



Dropped calls and overage barely move churn. Customers who call care churn less (45% vs 52%): they are engaged, not lost.

Share of leavers vs share of lost revenue, by monthly bill



$100+ customers are 11% of leavers but 29% of lost revenue, so we rank by revenue at risk, not risk alone.

Source: Company A data, 100,000 customers. Lost revenue: monthly bills of customers who left.

The core problem

Company A loses customers at predictable moments, but cannot yet see who is at risk, why, or who is worth saving.

65%
churn in month 11 of service, against 29% in month 9

64%
churn after monthly minutes fall by more than half

–$27K
lost per campaign when offers go to a random 3.5% of customers

Analytics task

Rank every customer by the chance of leaving (binary classification), judged by ROC-AUC, because a limited budget must reach the riskiest first.

Model

LightGBM ranks likely leavers best of four models

Model	ROC-AUC	Accuracy	F1	Leavers found in top 20%
Logistic regression	0.642	0.603	0.594	26.7%
Random forest	0.689	0.633	0.635	29.2%
XGBoost (tutorial settings)	0.703	0.644	0.642	30.0%
LightGBM (chosen)	0.706	0.647	0.645	30.2%

Accuracy and F1 use a 50% cut-off; ROC-AUC needs none. The tutorial baseline (XGBoost, 70/30 split) reached accuracy 0.637.

Why we chose LightGBM

Best at ranking: the highest ROC-AUC, which is what a limited call budget needs.

Built for messy data: handles the 43 variables with gaps and the text variables directly [5].

Explainable: every prediction splits into drivers (SHAP [6]), so each customer gets a reason.

Fair by design: ethnicity is never used; it would add only 0.0015 AUC (0.7079 with it).

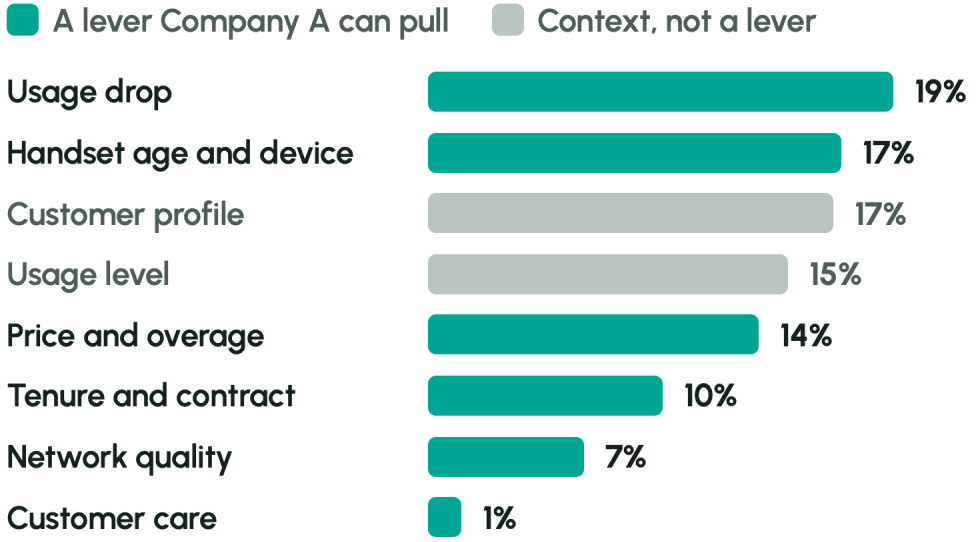
What 0.706 means Take one leaver and one stayer at random: the model rates the leaver riskier 71% of the time (a coin flip scores 50%). Of the top 20% it flags, 75% really leave, vs 50% on a random list.

5-fold stratified cross-validation on 100,000 customers; LightGBM varied by ±0.002 across folds

What the model learned

What drives the model's churn predictions

Share of the model's explanation, by driver family



Measured with SHAP [6] on customers the model had not seen. Customer profile: credit class, area, length of residence.

1. Minutes trend

Risk climbs once minutes fall about 30% below the customer's 3-month average.

2. Handset age

Risk jumps when the handset passes about 300 days, or 10 months.

3. Months in service

Risk peaks in months 10 to 12 of service, then fades.

On its own, the model found the same first-year cliff we saw in the raw data.

Source: LightGBM out-of-fold SHAP values, 100,000 customers; top 3 single drivers shown on the right

Proposal

A retention engine that acts on the reason

1. Score
Every customer, every month, with the LightGBM model

2. Explain
Name each customer's main reason from the SHAP drivers

3. Prioritise
Rank by churn risk × monthly bill; contact the top 3.5%

4. Act and learn
Matched offer, a held-out control group, monthly retraining

Playbook for the top 20% risk group (20,000 customers; 75% of them left)

Main reason	Share	Churn	Avg bill	At risk / month	Action
Usage drop	49%	80%	$45	$343K	Check-in call + usage bonus
Tenure and contract	21%	71%	$61	$182K	Renewal offer at month 9–10
Handset age	15%	68%	$50	$105K	Early handset-upgrade offer
Price and overage	13%	70%	$87	$153K	Right-size plan, end overage
Network quality	2%	66%	$61	$17K	Network fix + service credit

At risk = churn probability × monthly bill. Customer care (4 customers) not shown.

Business impact

Targeting by revenue at risk makes retention pay

One monthly campaign, per 100,000 customers



3,501
customers called, the top 3.5%

50
customers kept who would have left

70%
return: $68K kept for $40K cost

Run monthly: about $338K a year per 100,000 customers, preventing 2.5% of all churn (leaders reach up to 15% [4]).

Profitable in every scenario tested

By save rate and offer value, at 2% churn

Save rate	$30 offer	$60 offer	$90 offer
15%	$13K	$8K	$5K
25%	$47K	$28K base	$19K
35%	$100K	$65K	$43K

By real churn rate and save rate, $60 offer

Churn rate	Save 15%	Save 25%	Save 35%
1%	$0.3K	$5K	$12K
2%	$8K	$28K base	$65K
3%	$22K	$75K	$157K

Base case: 2% churn [1][2], $5 per call [7], $60 offer, 25% of reached leavers kept, recalibrated risks [8]

Roll-out

Prove it in a 3-month pilot, then scale

Months 1–3

Pilot

- Score all customers every month
- Call the top 3.5% by revenue at risk with the matched offer
- Hold back 20% of them as a control group

Months 4–6

Scale

- Roll out if called customers churn less than the control group and ROI is positive
- Tune offer value and contact share with the measured save rate

Months 7–12

Embed

- Automate monthly scores and reason codes
- Month-9 renewal offer before every first anniversary
- Retrain quarterly and review fairness

How we will measure success

Churn of called customers vs control (main KPI)

Revenue kept per dollar spent

Offer take-up by customers who would have stayed

Model ROC-AUC on each new month

Pilot budget: about $40K a month per 100,000 customers for calls and offers, expected to be repaid by the revenue kept.

Control group: flagged by the model but not called, so the gap between the groups shows the real effect

Risks and ethics

What could go wrong, and how we manage it

Offers may not change minds

The model finds who will leave, not who can be persuaded. The control group measures the real effect before we scale.

Paying people who would stay

We rank by risk and value, budget for 10% take-up among loyal customers, and track take-up every month.

Unfair treatment

Ethnicity is never used; it would add only 0.0015 AUC. Offer rates by area and credit class are reviewed every quarter.

Customer privacy

Only data Company A already holds to provide service. Opt-outs are respected and no customer data leaves the company.

The model goes stale

ROC-AUC is checked on every new month; the model is retrained quarterly, or sooner if accuracy drops.

A study sample, not the real base

The data is a 50/50 sample. Risks are recalibrated to real churn levels [8] and results are tested at 1% to 3% churn.

Fairness check: LightGBM AUC 0.7079 with ethnicity vs 0.7064 without it

Sources

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